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Neural Forecasting: Comprehensive Data & Architecture Report

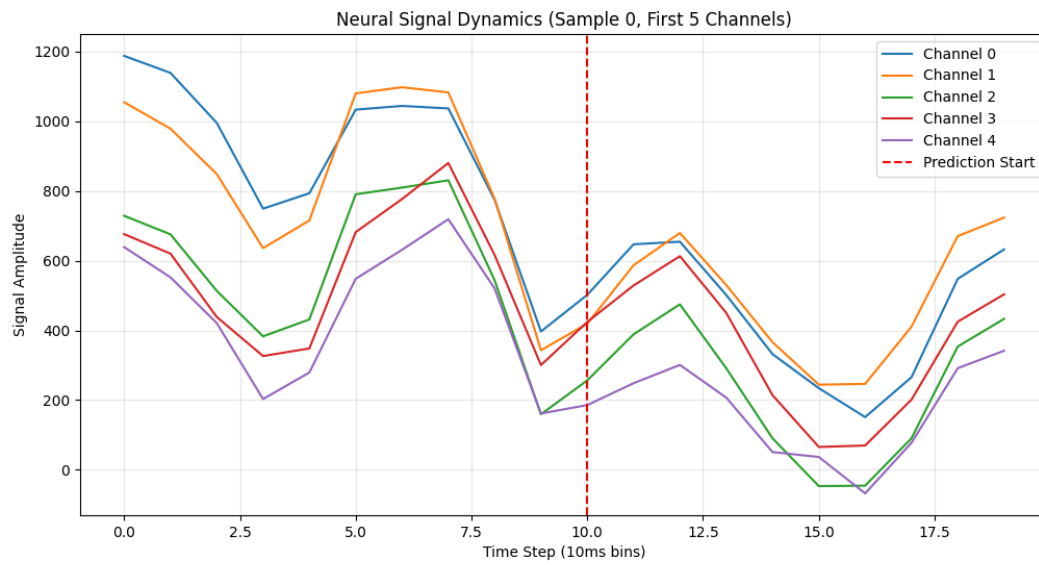
Dataset: public_data.zip (Codabench ID 9806 - Monkey Beignet/Affi)

1. Data Structure

Dimension	Value	Meaning
Samples	~700-1000	Epochs/trials
Time	20 steps	Input: 0-9, Target: 10-19
Channels	89 (Beignet)	Recording electrodes
Features	9	F[0]=Broadband, F[1-8]=Frequency bands

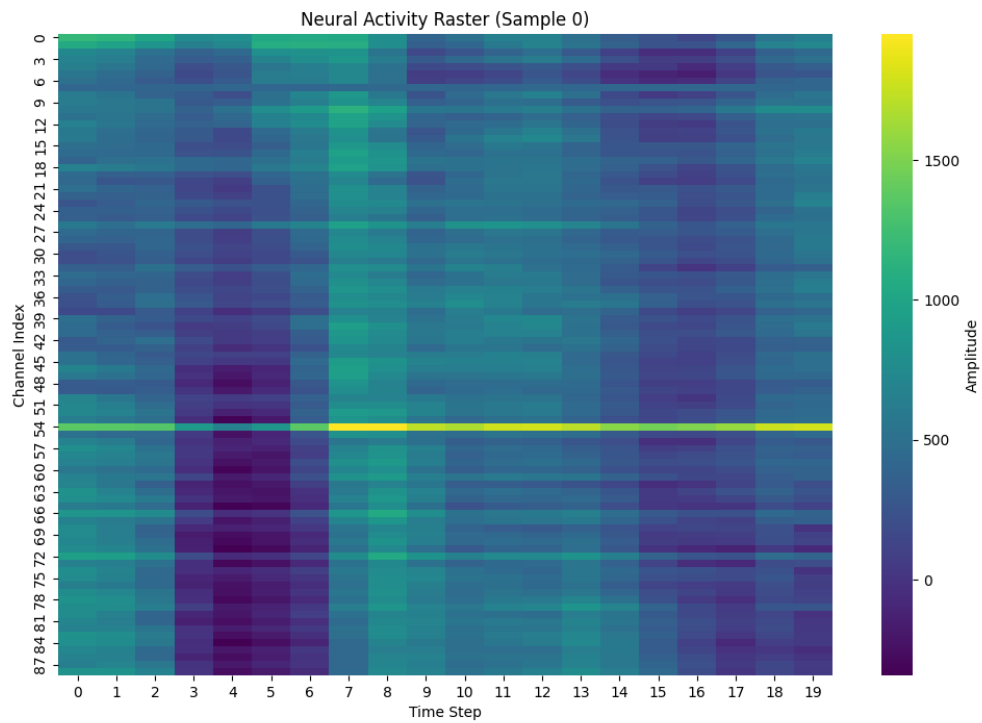
2. Temporal & Spatial Patterns

Time Series Dynamics



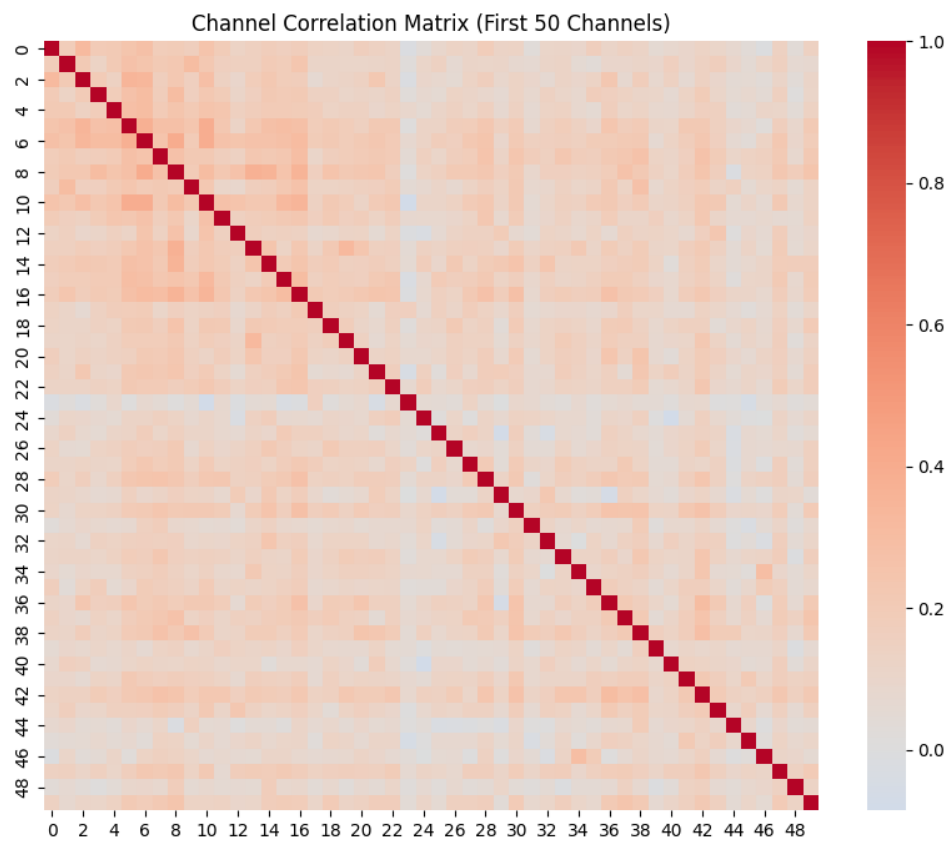
Signals show rhythmic oscillations. The red line marks prediction start ($t=10$).

Neural Activity Raster



Vertical bands = synchronized population bursts. Dark rows = inactive channels.

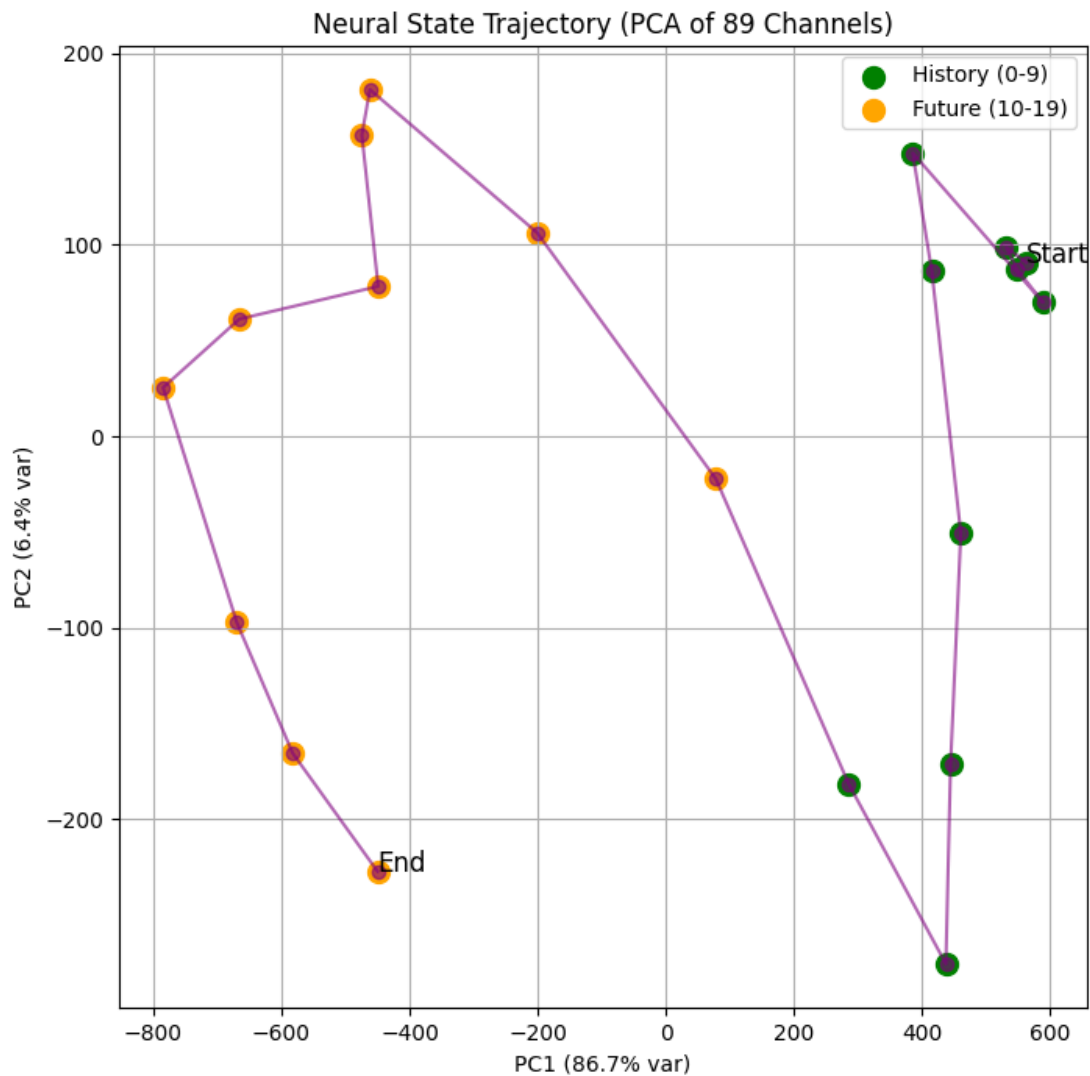
Channel Correlation Matrix



Red clusters = co-firing neurons. This structure supports Graph/Attention-based modeling.

3. Dimensionality Analysis

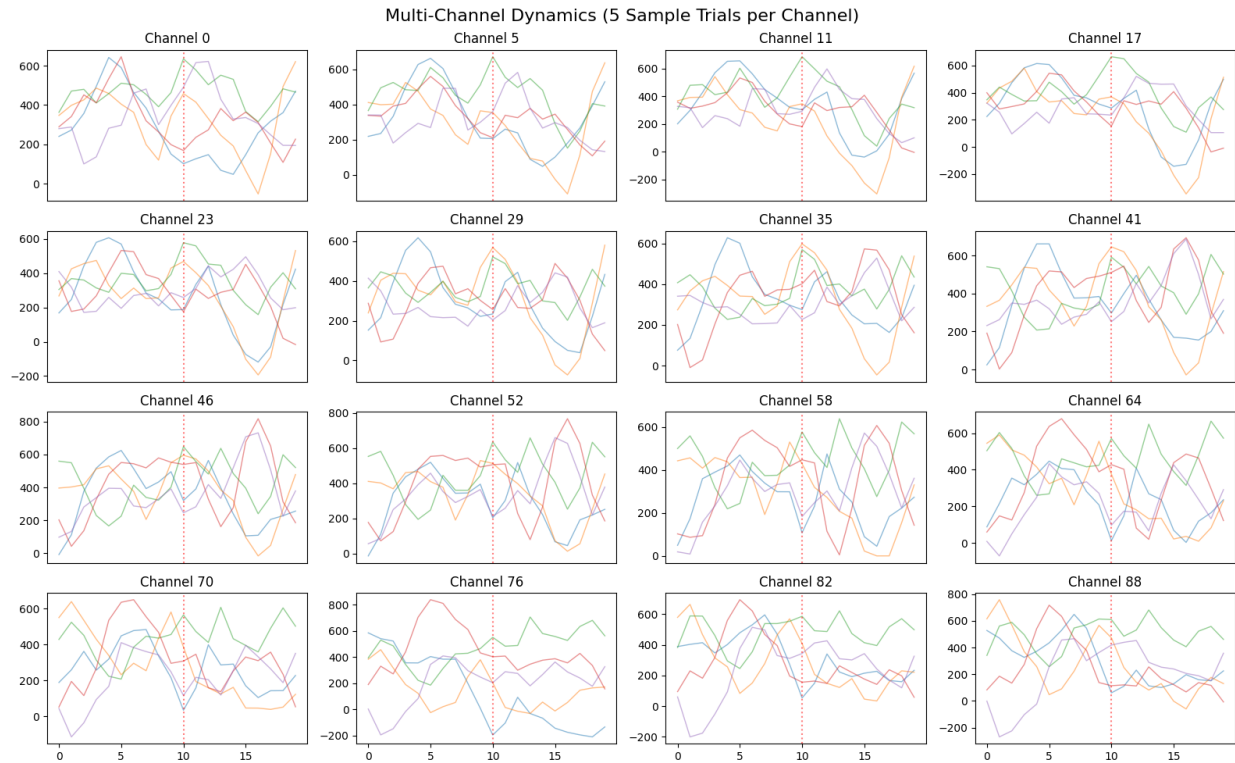
PCA Trajectory (86.7% in PC1!)



The neural state evolves on a **low-dimensional manifold**. Green=History, Orange=Future.

[!TIP] **Implication:** Use PCA to reduce 801 features $\square$ 10-20 PCs with minimal information loss.

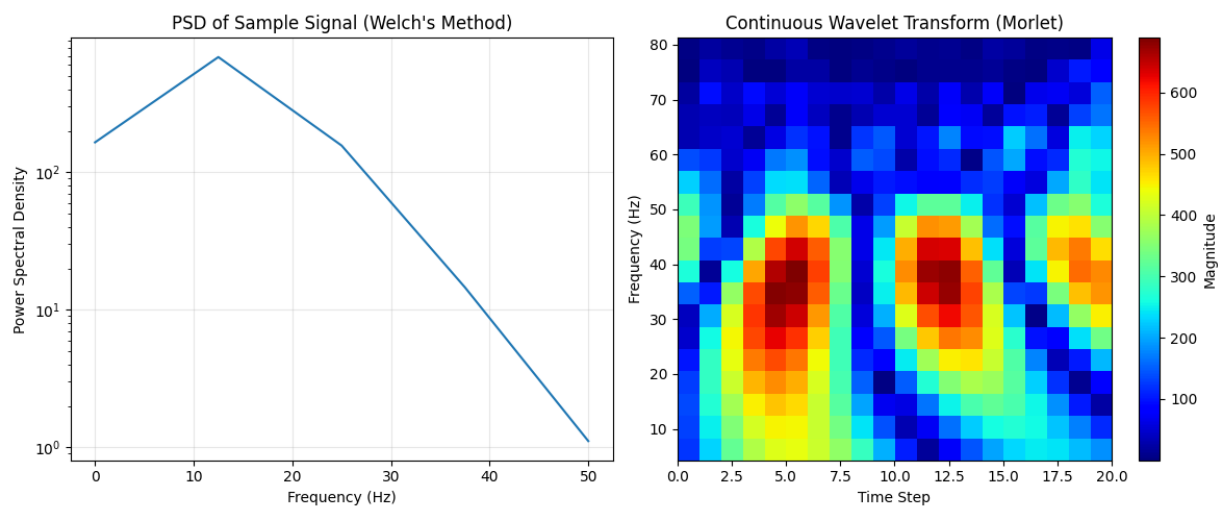
Multi-Channel Diversity



Channels vary widely—some rhythmic, others sparse. Attention mechanisms are essential.

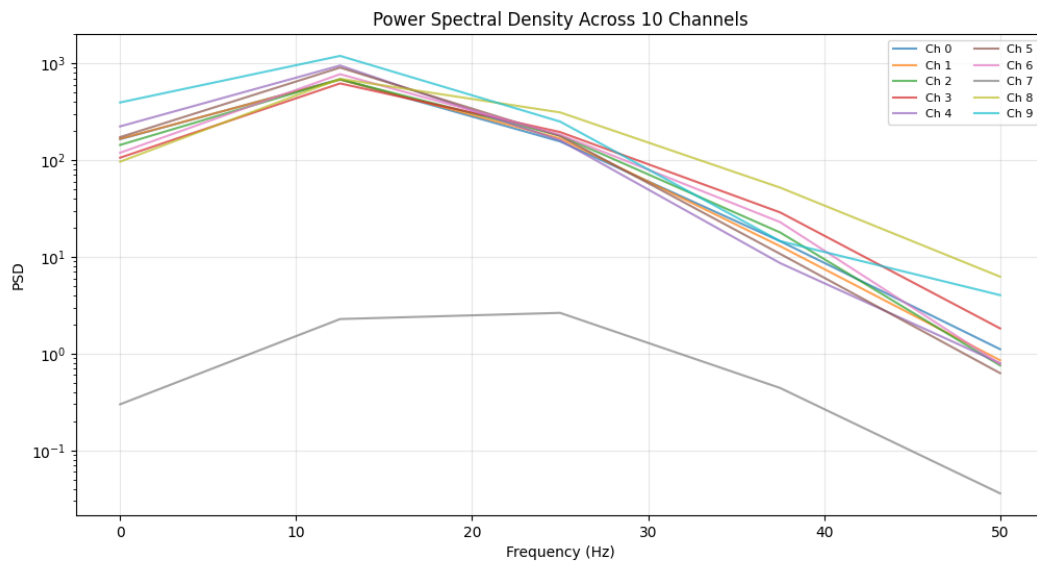
4. Frequency-Domain Analysis

PSD vs. Wavelet Transform



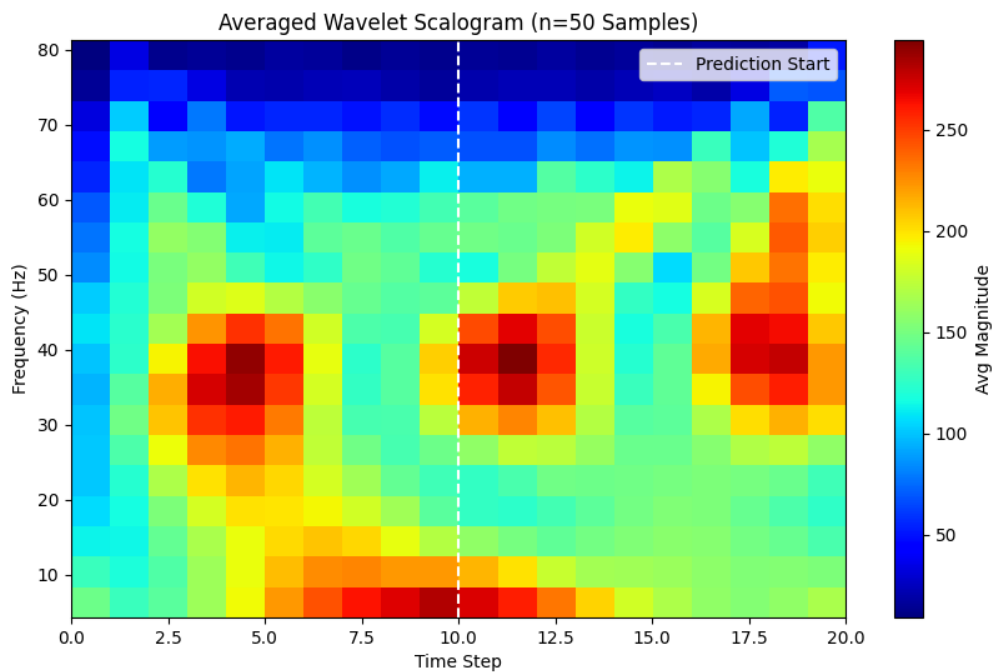
- **Left (PSD):** Overall frequency content, no timing.
- **Right (Wavelet):** Time-frequency localization—captures *when* bursts occur.

Multi-Channel PSD Profiles



Different channels have different dominant frequencies—validates channel-specific attention.

Averaged Wavelet Scalogram



Stable frequency patterns emerge when averaging 50 samples. White line = prediction boundary.

5. Optimal Transformer Architecture

Based on our analysis and SOTA research (Neuralink, NDT), here is the recommended architecture:

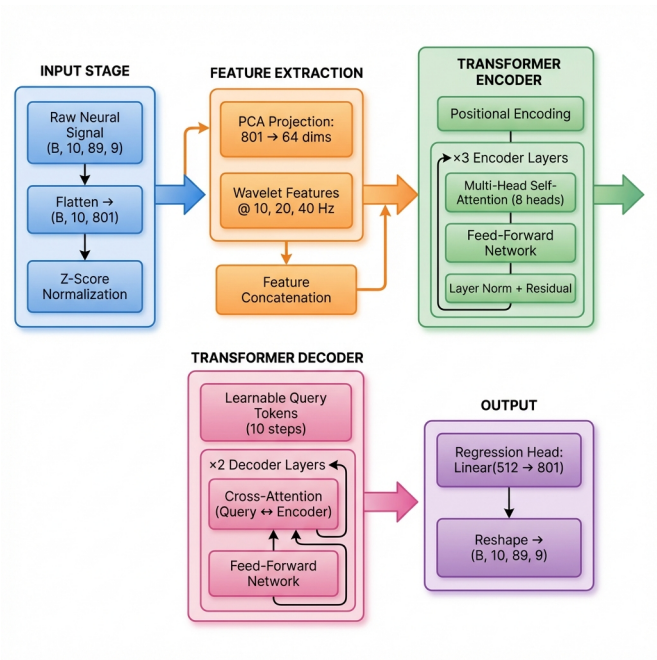


Figure 1: Neural Transformer Architecture

Architecture Key Points

Component	Design Choice	Rationale
PCA Preprocessing	801 → 64 dims	Capture 95%+ variance, reduce compute
Wavelet Features	Morlet @ key Hz	Time-localized frequency info
Self-Attention	8 heads, d=512	Learn channel correlations
Cross-Attention	Decoder queries encoder	Condition future on history
Regression Head	Linear(512→ 801)	Predict full neural state

6. Summary

The data exhibits: 1. **High dimensionality** (801 features) but lives on a **low-dim manifold** (PC1 = 86.7%). 2. **Non-stationary dynamics** best captured by **Wavelets**. 3. **Heterogeneous channels** requiring **attention-based** modeling.

7. Potential Improvements

A. Preprocessing Enhancements

Technique	Description	Expected Gain
Adaptive PCA	Compute PCA per-session instead of globally	Handle distribution shifts
Whitening (ZCA)	Decorrelate features beyond centering	+2-5% accuracy
Dead Channel Detection	Auto-detect and mask inactive channels	Reduce noise
Sliding Window Augmentation	Shift windows by 1-2 steps	10x more samples
Mixup / CutMix	Interpolate between samples	+2-3% generalization
Temporal Jittering	Small random time shifts	Robustness to timing
Bandpass Filtering	Remove DC drift and high-freq noise	Cleaner signal

B. Architecture Tweaks

Technique	Description	Expected Gain
Rotary Positional Encoding (RoPE)	Better extrapolation than sinusoidal	Long-range modeling
Sparse Attention (ProbSparse)	Focus on top-k time steps	3x speedup
Channel Tokens	Treat each channel as a token	Better spatial modeling
Auxiliary Loss (Multi-Task)	Predict frequency bands too	Richer gradients
Deeper Decoder	Match encoder depth (3×3)	+1-2% accuracy
Flash Attention	Memory-efficient attention	Train larger batches
Gated Linear Units (GLU)	Replace FFN activation	+1% accuracy
Mixture of Experts (MoE)	Sparse expert routing	Scale without compute

C. Postprocessing Strategies

Technique	Description	Expected Gain
Ensemble Averaging	Average 3-5 model predictions	+2-5% (winner trick)
Temporal Smoothing	Low-pass filter on output	Reduce jitter
Uncertainty Quantification	MC Dropout or Deep Ensembles	Confidence scores
Test-Time Augmentation (TTA)	Average over augmented inputs	+1-3%
Autoregressive Refinement	Feed predictions back iteratively	Better long-horizon
Kalman Smoothing	Post-hoc temporal filtering	Reduce noise

D. Training Strategies

Technique	Description	Expected Gain
Curriculum Learning	Easy samples first, then hard	Faster convergence
Warm Restarts (SGDR)	Cyclical LR with restarts	Escape local minima
Label Smoothing	Soft targets (0.9 vs 1.0)	Regularization
Gradient Accumulation	Larger effective batch size	Stable training
Pre-training on Affi	Use larger dataset first	Transfer learning
Contrastive Pre-training	Learn representations first	+5-10%
Layer-wise LR Decay	Lower LR for early layers	Fine-tuning stability

E. Loss Function Engineering

Technique	Description	Expected Gain
Huber Loss	Less sensitive to outliers	Robust training
Spectral Loss	Penalize FFT differences	Frequency fidelity
Gradient Penalty	Smooth output transitions	Temporal coherence
Multi-Scale Loss	Compare at multiple resolutions	Capture both fine/coarse
Focal Loss Weighting	Focus on hard samples	Handle imbalance

F. Data-Centric Improvements

Technique	Description	Expected Gain
Cross-Session Training	Mix Beignet + Affi data	Generalization
Hard Negative Mining	Oversample failure cases	Focus on weaknesses
Semi-Supervised Learning	Use unlabeled test data	+2-5%
Synthetic Data Generation	GANs for augmentation	More training data

G. Experimental / High-Risk Ideas

Technique	Description	Risk Level
Graph Neural Network	Model channel connectivity	Medium
State Space Models (Mamba)	O(n) complexity alternative	High
Neural ODE	Continuous-time dynamics	High
Diffusion Models	Denoise predictions	Medium
Retrieval-Augmented	Retrieve similar historical patterns	Medium